Sofia Velasquez Sierra

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Education

Carnegie Mellon University | M.S. Biomedical Engineering (Computational and Machine Learning Track) August 2026

McGill University | B.Eng. Computer Engineering June 2025

Experience

Weber Neural Interfaces Lab, Carnegie Mellon University | Research Assistant April 2026 - Present

- Build real-time pipelines that decode sEMG signals from Meta's wristband prototype, working alongside researchers and Meta engineers; defined 4 feasibility criteria Meta adopted to gate hardware decisions

BTG Pactual | Software Engineering Intern | Sao Paulo, Brazil June 2025 - August 2025

- Built 3 internal tools in React, TypeScript and C#/.NET for 15+ credit analysts after agreeing requirements with 4 teams; all 15+ used them daily within 2 weeks, cutting reporting time 40%

BNP Paribas | IT Production Security Intern | Montreal, Canada May 2024 - August 2024

- Replaced 5 manual tools with one monitoring system for 8 analysts, raising event coverage from 85% to 97%
- Wrote the triage framework for 200+ monthly events and documented the system for non-engineers; adopted as the team standard within 3 weeks

Projects

Rollout | Counterfactual debugger for AI agents | Next.js, TypeScript, three.js, Claude API October 2026

- Designed and built an interactive tool for debugging AI agents in a simulated world: scrub a run, fork it at any frame, redirect the agent in plain language, and compare outcomes side by side
- Built a real-time 3D renderer in three.js with realistic materials, soft shadows and a tilt-shift camera; each branch is its own robot test bay, with labels that track positions in the scene
- Claude Haiku drives the agent with streamed reasoning while Claude Sonnet scores each branch separately against a fixed rubric; server-side calls with input validation, per-IP rate limits, token caps and a rule-based fallback
- Cut wall collisions from 74-85 per run to 3 in total across 8 recorded branches by giving the agent pre-computed routes, so it chooses targets and risk rather than steering square by square
- Deterministic simulator with an append-only event log: any frame replays exactly and a fork is a pointer, not a copy; 68 automated tests
- Four full design iterations, from an instrument-panel dashboard to an immersive "parallel worlds" interface; keyboard control throughout, reduced-motion support, a 2D fallback and layouts down to tablet width

AgentTrace | AI agent evaluation tool | Python, Claude API May 2026

- Split simulation from judgment: one model plays the agent in a live streaming chat, a second scores the conversation on helpfulness, policy compliance, tone and failure modes and returns 3 concrete fixes per run

NeuroBridge | Wearable EMG system for stroke rehabilitation | Swift, SwiftUI, Swift Charts Spring 2026

- Designed and built the iOS app: multi-series muscle-activation charts filterable by muscle group, a raw waveform view, and AI exercise suggestions that wait for clinician approval before reaching the patient

MerchantPulse | Customer insight tool | Python, Claude API, Streamlit April 2026

- Interviewed 3 business owners before building, then iterated on the output format until all 3 could act on it without explanation

ROF | Web platform | TypeScript, React, Next.js, tRPC, Prisma, Tailwind December 2024

- Designed and shipped a 9-page platform with onboarding, ID verification and an analytics dashboard, replacing Excel and Word workflows for real users

Skills

Front end: TypeScript, React, Next.js, three.js, CSS, Tailwind, animation, accessibility, Swift and SwiftUI
AI and agents: Claude API (Haiku, Sonnet), streaming interfaces, agent evaluation, Claude Code, Cursor
Also: Python, PostgreSQL, Node.js, tRPC, Prisma, Git, Vercel; English, French, Spanish, Portuguese